

NAME: - Madhu M M

WEEK 5 CASE STUDY: - Smart Manufacturing Data Analytics
Platform

Project Overview

The objective of this project is to ingest IoT sensor data, perform transformations, and validate the data using data quality checks with PyDeequ. The implementation has been done in two ways:

1. Databricks Jobs using DBFS storage.
2. ADF Pipeline using Azure Storage Account.

The project demonstrates orchestration, automation, and data quality validation for structured IoT data. Data Description the IoT dataset contains information about manufacturing plants and sensor readings. It includes a PlantID to identify the plant, a DeviceID to uniquely identify each IoT sensor, and a Timestamp representing the UTC time of the reading. Sensor measurements include Temperature in Celsius, Humidity in percentage, Vibration speed in mm/s, Pressure in bar, and Energy consumption in kWh. The Status column indicates operational status with values like OK, WARNING, or ALERT.

Project Flow:

Step 1: Upload Data to DBFS CSV files were uploaded to DBFS using the Databricks UI.

It contains three plants **data plant 1, plant 2 and plant 3.**

Example

path:

`f"dbfs:/FileStore/shared_uploads/traininguser8@sudosu.ai/{plant-01}.csv"`

Data was read using Spark DataFrame

Step 2: Notebook Development to handle:

1. Data ingestion: Data ingestion through parameterized notebook based on the plant name data will be fetch from the storage and the process the data

2. Data transformation: Applied the data transformation to filter

3. Data quality validation using PyDeequ

4. Report generation Widgets were used to pass parameters, allowing notebooks to process one file at a time.

Step 3: Data Quality Checks with PyDeequ Installed Scala runtime and Deequ JAR libraries on the cluster.

Defined rules like:

- Completeness of columns (PlantID, DeviceID) .
- Non-negative numeric values.
- Allowed values for Status column (OK, WARNING, ALERT) Results were written as JSON to DBFS.

Step 4: Job Orchestration

Created the Databricks job linking the ingest data and data quality check.

Jobs were executed manually and automated through GitHub Actions using a YAML workflow file.

Output included Deequ JSON reports for validation.

Step 5: Observations

- Databricks jobs process one file at a time via widgets.
- DBFS storage is suitable for development and prototyping.
- **GitHub Actions:** workflow successfully triggers the job but requires correct cluster setup and library installation.

Supported Snapshots:

The screenshot displays the Databricks web interface. The top navigation bar includes the Microsoft Azure logo, the Databricks logo, and a search bar. The left sidebar shows the workspace structure with options like New, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Data Engineering, Job Runs, AI/ML, Playground, Experiments, Features, Models, and Serving. The main area shows an 'Untitled Notebook' with a code cell containing a warning message: 'UserWarning: DataFrame constructor is internal. Do not directly use it.' Below the code cell, a table is displayed with 7 rows and 3 columns: 'heck_level', 'check_status', and 'constraint'. The table shows various compliance constraints and their statuses. The bottom status bar indicates '7 rows | 0.90s runtime' and 'Refreshed now'.

heck_level	check_status	constraint
1	Success	SizeConstraint(Size(None))
2	Success	ComplianceConstraint(Compliance(Temperature_C is positive, COALESCE(CAST(Temperature_C AS DECIMAL(20,10)), 1.0) >
3	Success	ComplianceConstraint(Compliance(Temperature_C is non-negative, COALESCE(CAST(Temperature_C AS DECIMAL(20,10)), 0.0) >
4	Success	ComplianceConstraint(Compliance(Humidity_pct is non-negative, COALESCE(CAST(Humidity_pct AS DECIMAL(20,10)), 0.0) >
5	Success	ComplianceConstraint(Compliance(Vibration_mm_s is non-negative, COALESCE(CAST(Vibration_mm_s AS DECIMAL(20,10)), 0.0) >
6	Success	ComplianceConstraint(Compliance(Pressure_bar is non-negative, COALESCE(CAST(Pressure_bar AS DECIMAL(20,10)), 0.0) >
7	Success	ComplianceConstraint(Compliance(Pressure_bar is non-negative, COALESCE(CAST(Pressure_bar AS DECIMAL(20,10)), 0.0) >

Microsoft Azure databricks

Search data, notebooks, recents, and more... CTRL + P

+ New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Data Engineering

Job Runs

AI/ML

Playground

Experiments

Features

Models

Serving

File Edit View Run Help Python Tabs: ON ☆ Last edit was 1 minute ago Run all Training User6's Cluster... Schedule (1) Share

Plant_ID

plant-01

```
0)), 0.0) >= 0, None)))"
},
{
  "check_status": "Success",
  "check_level": "Error",
  "constraint_status": "Success",
  "check": "IoT Data Quality Checks",
  "constraint_message": "",
  "constraint": "ComplianceConstraint(Compliance(Pressure_bar is non-negative, COALESCE(CAST(Pressure_bar AS DECIMAL(20,10))),
0.0) >= 0, None)))"
},
{
  "check_status": "Success",
  "check_level": "Error",
  "constraint_status": "Success",
  "check": "IoT Data Quality Checks",
  "constraint_message": "",
  "constraint": "ComplianceConstraint(Compliance(Energy_kwh is non-negative, COALESCE(CAST(Energy_kwh AS DECIMAL(20,10))), 0.
0) >= 0, None)))"
}
```

25°C Mostly cloudy

Quick search

ENG IN

14:23 15-08-2025

raw/plant-01.txt - Mic Job runs - Databricks madhum0089/wee Explain data cleaning Maven Central: com Maven Repository: co

adb-361337280981643.3.azuredatabricks.net/jobs/runs?o=361337280981643

Microsoft Azure databricks

Search data, notebooks, recents, and more... CTRL + P

databricks-madhu

+ New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Data Engineering

Job Runs

AI/ML

Playground

Experiments

Features

Models

Serving

Jobs & Pipelines

Create new

Ingestion pipeline
Ingest data from popular apps, databases and file sources

ETL pipeline
Build ETL pipelines using SQL and Python

Job
Orchestrate notebooks, pipelines, queries and more

Jobs & pipelines Job runs

Send feedback

1 1 0 14 Aug, 12 PM UserCanceled

13 Aug, 12 AM 13 Aug, 12 PM 14 Aug, 12 AM 14 Aug, 12 PM

Failed Skipped Succeeded

Start time	Job	Run as	Launched	Duration	Status	Error code	Run parameters
Aug 14, 2025, 05:56 PM	Untitled	Training User6	By runs submit API	1m 53s	Succeed...		Plant_ID: plant_01
Aug 14, 2025, 05:34 PM	New Job Aug 14, 20...	Training User6	Manually	21m 4s	Failed	RunExecutionError	
Aug 14, 2025, 05:39 PM	Untitled	Training User6	By runs submit API	35s	Failed	RunExecutionError	

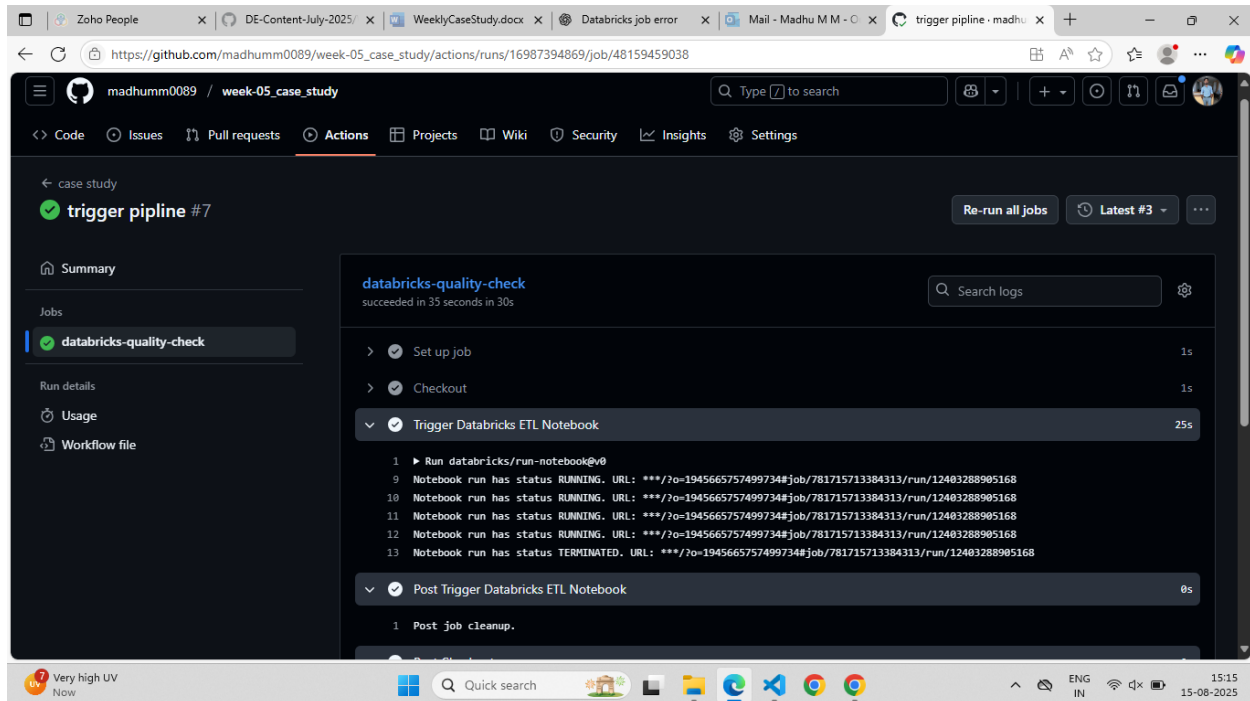
< Previous Next >

23°C Mostly cloudy

Search

ENG IN

18:04 14-08-2025



Method 2: ADF Pipeline

Step 1: Storage Setup Created an Azure Storage Account with three containers: raw – store raw CSV files

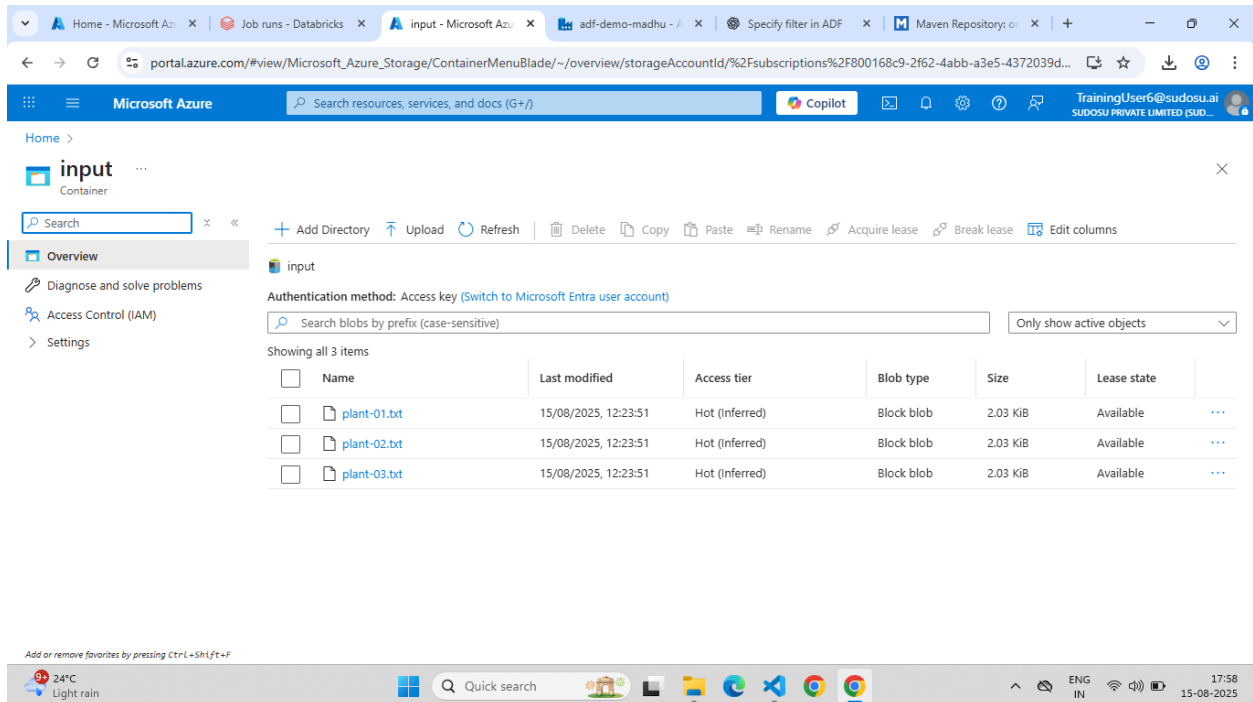
processed – store transformed data using dataflow

Step 2: Linked Services & Datasets Created Linked Services for the storage account and Databricks workspace. **Created Datasets:** Source: raw Sink: processed.

Step 3: Pipeline Design ADF pipeline consisted of:

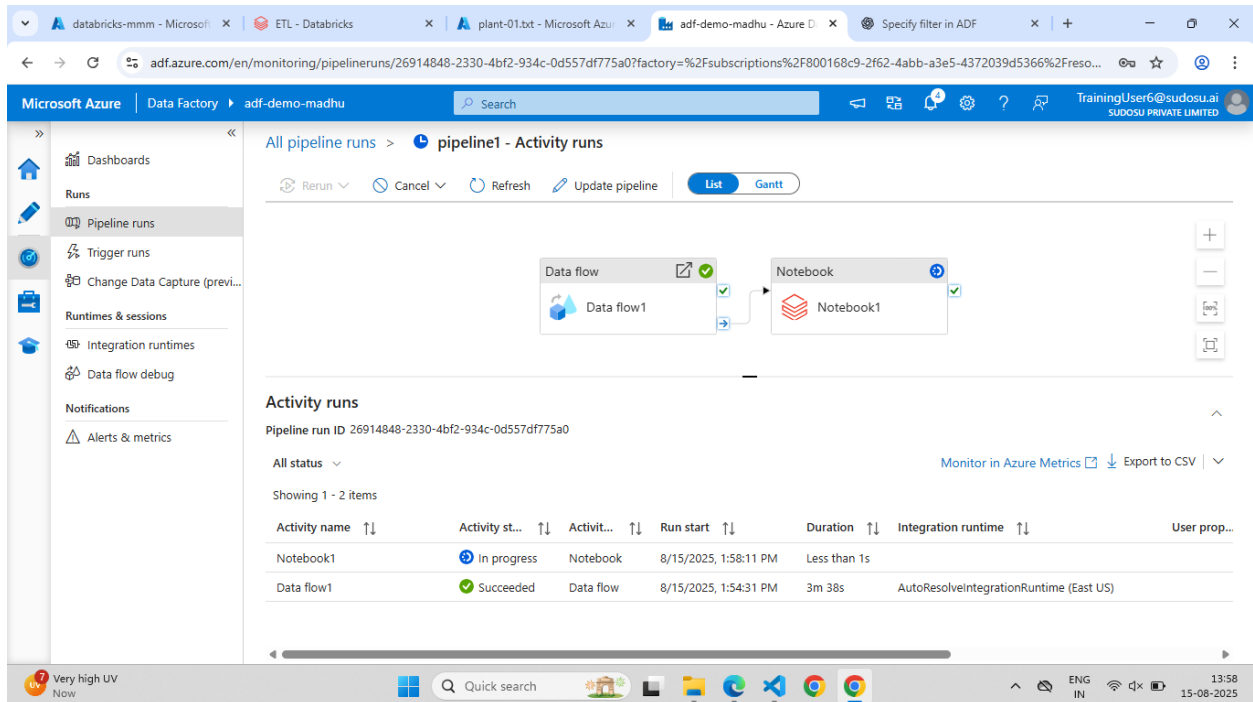
Dataflow: Moves data from raw → filters -> processed

Databricks Notebook Activity: Performs data quality checks on all files in processed data.



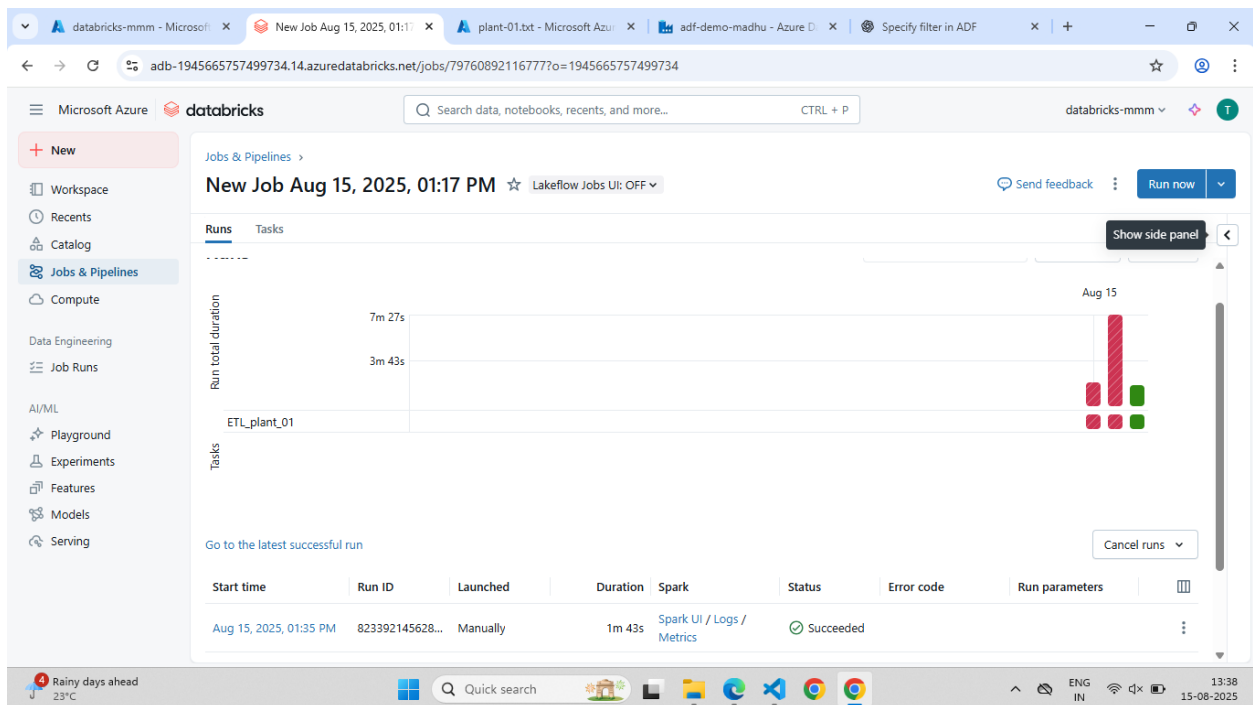
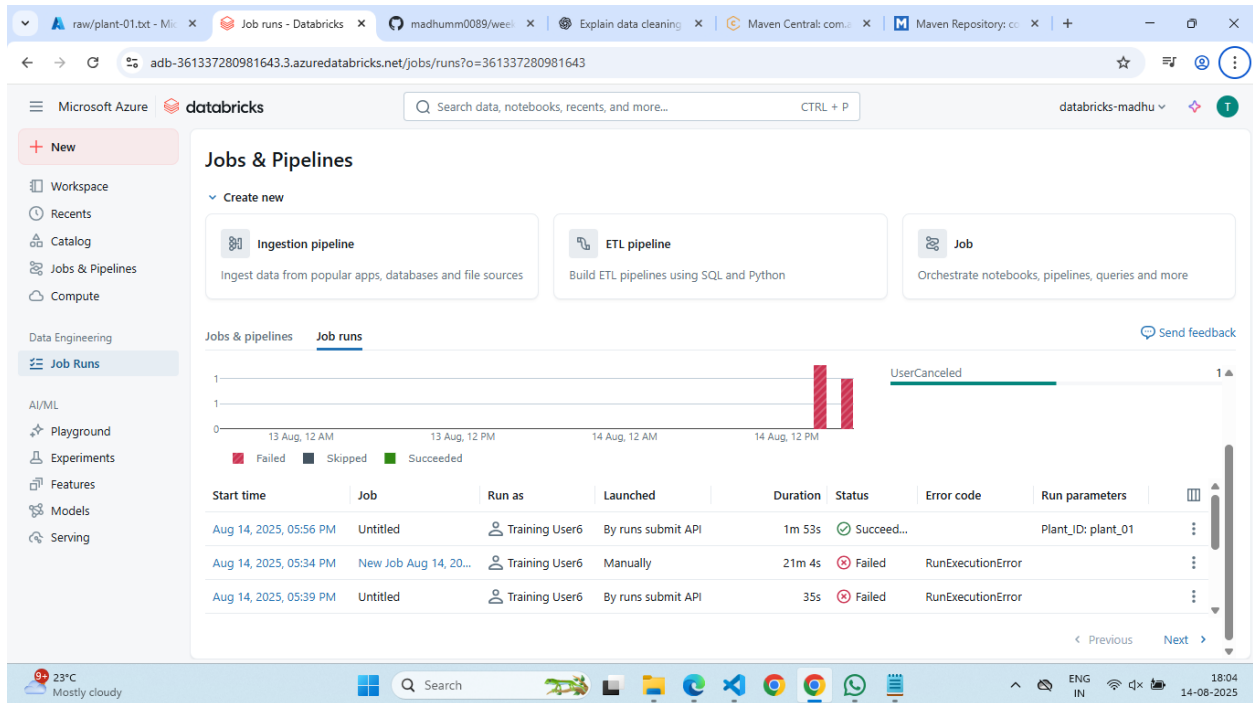
Microsoft Azure portal interface showing the 'input' container overview. The page displays a table of blobs with columns: Name, Last modified, Access tier, Blob type, Size, and Lease state. Three blobs are listed: plant-01.txt, plant-02.txt, and plant-03.txt, all with a size of 2.03 KiB and an 'Available' lease state.

Name	Last modified	Access tier	Blob type	Size	Lease state
plant-01.txt	15/08/2025, 12:23:51	Hot (Inferred)	Block blob	2.03 KiB	Available
plant-02.txt	15/08/2025, 12:23:51	Hot (Inferred)	Block blob	2.03 KiB	Available
plant-03.txt	15/08/2025, 12:23:51	Hot (Inferred)	Block blob	2.03 KiB	Available



Microsoft Azure Data Factory portal interface showing the 'pipeline1 - Activity runs' view. The page displays a table of activity runs with columns: Activity name, Activity status, Activity type, Run start, Duration, Integration runtime, and User properties. Two activity runs are listed: Notebook1 (In progress) and Data flow1 (Succeeded).

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User properties
Notebook1	In progress	Notebook	8/15/2025, 1:58:11 PM	Less than 1s		
Data flow1	Succeeded	Data flow	8/15/2025, 1:54:31 PM	3m 38s	AutoResolveIntegrationRuntime (East US)	



Note: I have attached notebook and github actions yml file along with this documnet.